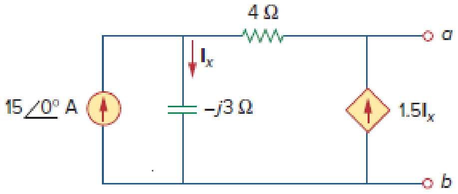
Chapter 10, Problem 61P

Problem

Find the Thevenin equivalent at terminals *a*-*b* of the circuit in Fig. 10.104.

Figure 10.104:



Step-by-step solution

Step 1 of 7

Refer to Figure 10.104 in the textbook.

Consider the following circuit to calculate the thevenin voltage:

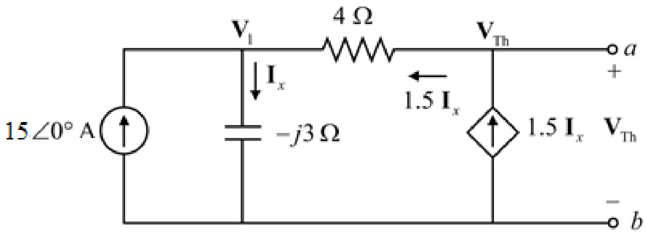


Figure 1

Step 2 of 7

Calculate the Thevenin voltage $\mathbf{V}_{\mathrm{Th}}$.

Apply Kirchhoff's current law at the node $\mathbf{V}_1$ for the circuit in Figure 1.

$$\begin{aligned}-15-1.5\mathbf{I}_x+\mathbf{I}_x&=0\\0.5\mathbf{I}_x&=-15\\\mathbf{I}_x&=-30\text{ A}\end{aligned}$$

Step 3 of 7

Calculate the voltage $\mathbf{V}_1$.

$$\mathbf{V}_1=\mathbf{I}_x(-j3)$$

Substitute -30 A for $\mathbf{I}_x$.

$$\begin{aligned}\mathbf{V}_1&=(-30)(-j3)\\&=j90\text{ V}\end{aligned}$$

Step 4 of 7

Consider the following current for the circuit in Figure 2.

$$\frac{\mathbf{V}_{\mathrm{Th}} - \mathbf{V}_1}{4} = 1.5\mathbf{I}_x$$
$$\mathbf{V}_{\mathrm{Th}} - \mathbf{V}_1 = 6\mathbf{I}_x$$
$$\mathbf{V}_{\mathrm{Th}} = 6\mathbf{I}_x + \mathbf{V}_1$$

Substitute $-30\ \mathrm{A}$ for $\mathbf{I}_x$ and $j90\ \mathrm{V}$ for $\mathbf{V}_1$.

$$\begin{aligned}\mathbf{V}_{\mathrm{Th}} &= 6(-30) + (j90) \\ &= -180 + j90\ \mathrm{V}\end{aligned}$$

Therefore, the Thevenin voltage $\mathbf{V}_{\mathrm{Th}}$ is $\boxed{-180 + j90\ \mathrm{V}}$.

Step 5 of 7

Consider the following circuit to calculate the thevenin impedance:

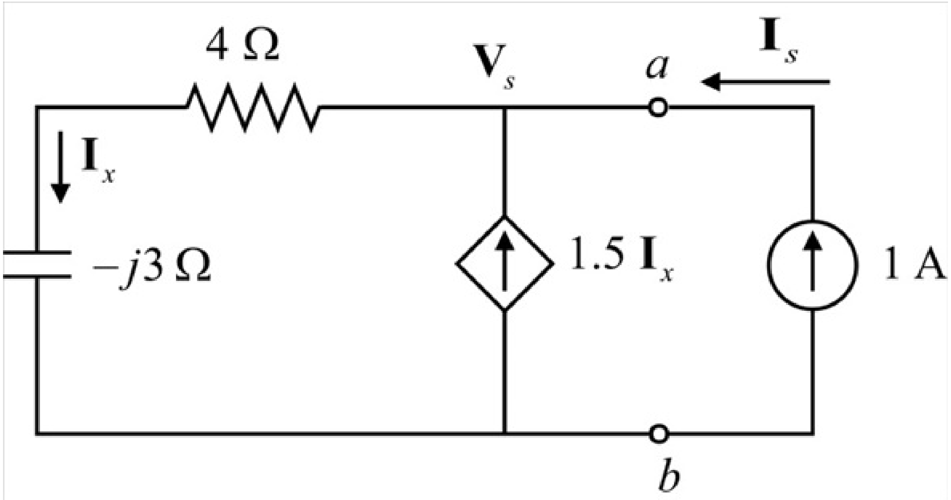


Figure 2

Step 6 of 7

Calculate the Thevenin impedance $\mathbf{Z}_{\text{Th}}$.

To calculate $\mathbf{Z}_{\text{Th}}$, we should open circuit all the independent current sources and short circuit all the independent voltage sources and consider a dummy current source $\mathbf{I}_s = 1\text{ A}$.

Apply Kirchhoff's current law at the node $\mathbf{V}_s$ for the circuit in Figure 3.

$$\mathbf{I}_s + 1.5\mathbf{I}_x - \mathbf{I}_x = 0$$
$$0.5\mathbf{I}_x = -\mathbf{I}_s$$

Substitute 1 A for $\mathbf{I}_s$ in $0.5\mathbf{I}_x = -\mathbf{I}_s$.

$$\begin{aligned}\mathbf{I}_x &= -\frac{1}{0.5} \\ &= -2\text{ A}\end{aligned}$$

Consider the following voltage for the circuit in Figure 3.

$$\mathbf{V}_s = (4 - j3)(\mathbf{I}_x)$$

Substitute -2 A for $\mathbf{I}_x$ in $\mathbf{V}_o = (4 - j3)(\mathbf{I}_x)$.

$$\begin{aligned}\mathbf{V}_s &= (4 - j3)(-2) \\ &= -8 + j6\text{ V}\end{aligned}$$

Calculate the Thevenin impedance $\mathbf{Z}_{\text{Th}}$.

$$\mathbf{Z}_{\text{Th}} = \frac{\mathbf{V}_s}{\mathbf{I}_s}$$

Substitute $-8 + j6\text{ V}$ for $\mathbf{V}_s$ and 1 A for $\mathbf{I}_s$ in $\mathbf{Z}_{\text{Th}}$.

$$\begin{aligned}\mathbf{Z}_{\text{Th}} &= \frac{-8 + j6}{1} \\ &= -8 + j6\ \Omega\end{aligned}$$

Therefore, the Thevenin impedance $\mathbf{Z}_{\text{Th}}$ is $\boxed{-8 + j6\ \Omega}$.

Step 7 of 7

Draw the Thevenin equivalent.

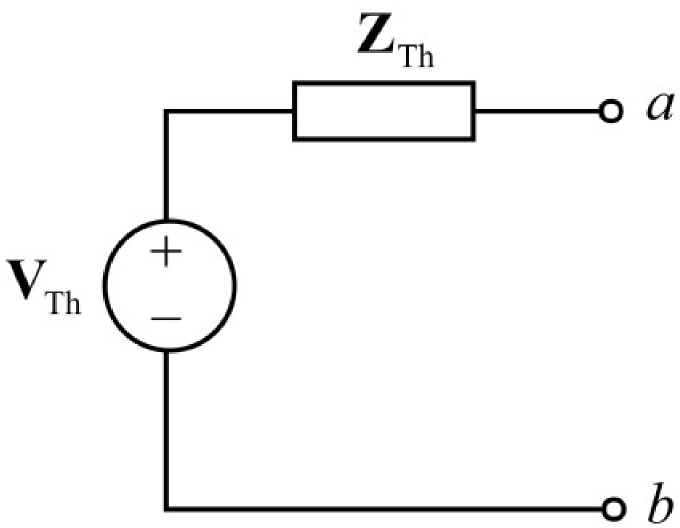


Figure 3